Chapter 12A - Ambulatory Cardiovascular Prevention and Headache

Hypertension, Hyperlipidemia, ASCVD Prevention, and Outpatient Headache Triage

This first ambulatory segment translates organ-system knowledge into the primary-care and clinic setting. The source chapter begins with hypertension, hyperlipidemia, and headache, so this segment focuses on outpatient cardiovascular risk management and headache triage. It is written to be clinically current: older source definitions are preserved as historical context when useful, but treatment thresholds and prevention frameworks are updated to modern guideline-style practice.

I. Ambulatory Cardiovascular Prevention Framework

Ambulatory cardiovascular medicine is not only “follow the blood pressure and cholesterol.” It is a repeated risk-management process: confirm the measurement, identify secondary or reversible contributors, estimate absolute risk, choose a medication strategy that matches comorbidities, and build a follow-up plan that detects both therapeutic failure and adverse effects. The clinic visit should answer four questions: Is the patient acutely unsafe? Is the abnormal value real? What is the patient’s absolute cardiovascular risk? What intervention gives the most benefit with the least burden?

The outpatient setting is also where longitudinal prevention happens. Hypertension, dyslipidemia, diabetes, smoking, obesity, chronic kidney disease, sleep apnea, sedentary behavior, and family history amplify each other. A small abnormality in one risk factor may matter more when several risks coexist. Conversely, a dramatic lab number may require urgent treatment even when the calculated 10-year risk appears low, such as LDL-C >=190 mg/dL or triglycerides >=500 mg/dL.

Clinic triage: when an outpatient cardiovascular problem is not outpatient anymore

Red flags that should push a clinic patient toward emergency or same-day high-acuity evaluation include chest pain concerning for acute coronary syndrome, focal neurologic deficit, hypertensive emergency with end-organ damage, acute pulmonary edema, syncope with exertion or abnormal ECG, suspected aortic dissection, papilledema, sudden severe headache, acute kidney injury with severe hyperkalemia, or severe medication adverse effect such as angioedema from an ACE inhibitor.

Ambulatory task	Practical action	Reason it matters
Confirm	Repeat BP/labs with correct technique; use home or ambulatory BP when white-coat or masked HTN is possible.	Prevents overtreatment of false positives and missed treatment of masked disease.
Risk-stratify	Estimate ASCVD risk, identify diabetes/CKD/ASCVD, ask about smoking, family history, pregnancy potential, and statin intolerance.	Medication thresholds are based on absolute risk, not isolated numbers alone.
Screen for secondary causes	Review medications/substances, renal disease, OSA symptoms, hypokalemia, abrupt onset, resistant HTN, very high TG/LDL.	Correctable causes may change the entire plan.
Treat + follow	Start lifestyle plus medication when indicated; schedule specific reassessment interval; define targets and monitoring labs.	Cardiovascular prevention fails when treatment has no measurable next step.

II. Hypertension in Ambulatory Medicine

Hypertension is one of the most common silent drivers of cardiovascular and renal morbidity. The source chapter emphasizes that most hypertension is essential, while secondary causes include renal/renovascular disease, endocrine disorders, medications, stimulants, coarctation, and obstructive sleep apnea. That framework remains clinically useful, but modern outpatient care places greater emphasis on accurate measurement, home or ambulatory confirmation, global ASCVD risk, and earlier treatment when risk is high.

Definitions and staging

Category	Blood pressure definition in adults	Clinic interpretation
Normal	SBP <120 and DBP <80 mm Hg	Reinforce healthy lifestyle and periodic recheck.
Elevated	SBP 120-129 and DBP <80	Lifestyle intervention; reassess because this stage often progresses.
Stage 1 HTN	SBP 130-139 or DBP 80-89	Lifestyle for all; add medication when ASCVD/CVD risk is high or BP remains above goal after lifestyle.
Stage 2 HTN	SBP >=140 or DBP >=90	Usually needs drug therapy plus lifestyle; many patients require 2 agents if >20/10 mm Hg above target.
Severe asymptomatic HTN	Often SBP >=180 or DBP >=120 without acute end-organ damage	Confirm, evaluate, intensify oral therapy, close follow-up; avoid rapid IV lowering unless emergency.
Hypertensive emergency	Severe BP elevation with acute end-organ injury	ED/ICU-level care; examples include encephalopathy, stroke, ACS, pulmonary edema, aortic dissection, AKI, papilledema, eclampsia.

Diagnosis requires repeated, accurate measurements rather than a single elevated reading, unless there is severe hypertension with clear end-organ injury. A practical clinic rule is to base the diagnosis on the average of at least two properly obtained readings on at least two occasions. Home blood pressure monitoring or 24-hour ambulatory blood pressure monitoring is especially useful when the office readings do not match the clinical picture, when white-coat hypertension is suspected, or when masked hypertension is possible.

Correct blood pressure measurement technique

• Use a validated device and the correct cuff size. A cuff that is too small can falsely elevate the reading; the bladder should encircle most of the arm circumference.

• Have the patient avoid caffeine, exercise, nicotine, and stimulant medications for about 30 minutes when possible before measurement.

• Seat the patient quietly for at least 5 minutes, with back supported, feet flat on the floor, legs uncrossed, and arm supported at heart level.

• Measure both arms at the first visit when feasible. Use the arm with the higher reading for future comparisons.

• Repeat the measurement if the value is unexpectedly high, and document whether pain, anxiety, withdrawal, missed medications, or acute illness may be contributing.

Target-organ injury and complications

The source chapter highlights the major target organs: heart, brain, kidney, retina, and vasculature. Chronic hypertension increases afterload, leading to concentric left ventricular hypertrophy, diastolic dysfunction, heart failure, myocardial infarction, and atrial fibrillation. Cerebrovascular complications include ischemic stroke, intracerebral hemorrhage, transient ischemic attack, and hypertensive encephalopathy. Renal disease develops through arteriolosclerosis, glomerular ischemia, albuminuria, declining GFR, and eventual chronic kidney disease. Retinal injury progresses from arteriolar narrowing and AV nicking to hemorrhages, cotton wool spots, exudates, and papilledema. Vascular consequences include peripheral arterial disease, abdominal aortic aneurysm, and aortic dissection risk.

Do-not-miss distinction: Hypertensive urgency is a severe number without acute injury; hypertensive emergency is severe pressure plus acute injury. Treat the patient and the organ injury, not the number alone. A patient with SBP 190 and no symptoms may need outpatient intensification; a patient with SBP 170 and acute pulmonary edema or neurologic deficit needs emergency management.

Initial outpatient evaluation of hypertension

Evaluation domain	What to assess	Examples / interpretation
History	Duration, prior readings, symptoms of end-organ injury, medication adherence, diet, alcohol, sleep apnea symptoms, pregnancy possibility.	Ask about NSAIDs, decongestants, stimulants, steroids, estrogen therapy, SNRIs, calcineurin inhibitors, cocaine/amphetamines, licorice.
Physical exam	Repeat BP, BMI/waist, cardiac exam, pulses, bruits, edema, fundoscopic clues when severe, neuro exam if symptomatic.	Unequal arm BP or pulse deficit may suggest vascular disease; abdominal bruit may suggest renovascular disease.
Baseline labs	BMP/creatinine/eGFR, potassium, sodium, calcium when indicated, fasting glucose or A1c, lipid panel, urinalysis, urine albumin-creatinine ratio.	Hypokalemia suggests primary aldosteronism or diuretic effect; albuminuria changes risk and drug selection.
Cardiac testing	ECG; echocardiography when symptoms, murmur, heart failure signs, or suspected LVH/structural disease.	LVH increases cardiovascular risk and supports aggressive risk-factor modification.
Secondary workup	Directed testing based on clinical clues.	Abrupt onset, resistant HTN, early age, hypokalemia, renal dysfunction after ACEi/ARB, OSA symptoms, paroxysms, or endocrine features.

Secondary hypertension pattern recognition

Pattern	Likely causes	Clues
Resistant HTN	OSA, primary aldosteronism, CKD, renal artery stenosis, medication/substance effect.	BP above goal on 3 drugs including a diuretic or controlled on 4 drugs. Confirm adherence and home BP first.
HTN + hypokalemia	Primary aldosteronism, diuretic use, Cushing syndrome, rare mineralocorticoid excess.	Suppressed renin with elevated aldosterone supports aldosteronism.
Abrupt worsening or ACEi/ARB creatinine rise	Renal artery stenosis, especially bilateral disease or solitary kidney.	Abdominal bruit, flash pulmonary edema, vascular disease.
Paroxysmal headache/palpitations/sweating	Pheochromocytoma/paraganglioma.	Episodic severe BP, tremor, pallor; test plasma free or urine fractionated metanephrines when suspected.
Young patient with upper-lower BP discrepancy	Coarctation of the aorta.	Arm HTN, weak/delayed femoral pulses, murmur.
Snoring/daytime sleepiness/obesity	Obstructive sleep apnea.	Morning headache, resistant HTN, nocturnal hypoxemia.

Lifestyle therapy: expected blood pressure effects

Lifestyle therapy is not a vague instruction to “do better.” It should be prescribed like a medication: dose, expected effect, follow-up, and barriers. Weight reduction, DASH-style eating, sodium reduction, aerobic activity, and alcohol moderation have additive effects and can lower systolic BP by clinically meaningful amounts.

Intervention	Practical prescription	Typical SBP effect
Weight loss	Aim for sustained loss in overweight/obesity; even modest loss matters.	Approximately 1 mm Hg per kg lost.
DASH diet	Fruits/vegetables, low-fat dairy, reduced saturated fat; high potassium unless contraindicated.	Often about 8-11 mm Hg in hypertension.
Sodium reduction	Target roughly <1.5-2.3 g sodium/day depending on patient context.	Often about 5-6 mm Hg.
Physical activity	At least 150 min/week moderate aerobic activity plus resistance training.	Often about 4-8 mm Hg.
Alcohol moderation	Men <=2 drinks/day; women <=1 drink/day; less if BP-sensitive.	Often about 4 mm Hg.

Medication strategy for hypertension

First-line drug classes for most nonpregnant adults include thiazide/thiazide-like diuretics, ACE inhibitors, ARBs, and long-acting dihydropyridine calcium channel blockers. Beta-blockers are not usually first-line for uncomplicated hypertension, but they are appropriate when there is a compelling indication such as coronary disease with angina, prior myocardial infarction, selected arrhythmias, heart failure with reduced ejection fraction, or certain pregnancy-related contexts using specific agents. Drug choice should consider diabetes, CKD with albuminuria, heart failure, gout, pregnancy potential, bradycardia, edema, hyperkalemia risk, and medication adherence.

Class	Clinical advantages	Important adverse effects / cautions
Thiazide/thiazide-like diuretics	Strong outcome data; chlorthalidone and indapamide have longer action; useful in salt-sensitive HTN and older adults.	Hypokalemia, hyponatremia, hyperuricemia/gout, hypercalcemia, photosensitivity; monitor electrolytes and renal function.
ACE inhibitors	Helpful in diabetes/CKD with albuminuria, CAD, HFrEF; reduce angiotensin II and aldosterone.	Cough, hyperkalemia, creatinine rise, angioedema, teratogenicity; avoid in pregnancy and bilateral renal artery stenosis.
ARBs	Similar renal/cardiac benefits to ACE inhibitors with less cough.	Hyperkalemia, creatinine rise, teratogenicity; do not combine routinely with ACE inhibitor.
Dihydropyridine CCBs	Amlodipine/nifedipine ER lower BP effectively; useful in older adults and isolated systolic HTN.	Peripheral edema, flushing, headache, gingival hyperplasia.
Non-DHP CCBs	Verapamil/diltiazem can help rate control.	Bradycardia, heart block, constipation; avoid or use caution in HFrEF and with beta-blockers.
Beta-blockers	Compelling indications include CAD/angina, post-MI, some arrhythmias, HFrEF with evidence-based agents.	Bradycardia, fatigue, bronchospasm risk, depression/sexual dysfunction; mask hypoglycemia symptoms.
Mineralocorticoid receptor antagonists	Useful in resistant HTN and primary aldosteronism; spironolactone often effective.	Hyperkalemia, renal dysfunction, gynecomastia with spironolactone.

Hypertension treatment algorithm

1. Confirm accurate BP and assess for emergency. 2. If elevated or stage 1 with low short-term risk, prescribe lifestyle and reassess. 3. If stage 1 with clinical CVD, diabetes/CKD, or estimated 10-year ASCVD risk >=10%, start medication plus lifestyle. 4. If stage 2, start medication plus lifestyle; consider two first-line agents from different classes when BP is >20/10 mm Hg above target. 5. Recheck in about 1 month after medication changes, sooner for severe readings or symptoms. 6. Monitor BMP 1-4 weeks after ACEi/ARB/diuretic initiation or dose increase. 7. If uncontrolled on 3 agents including a diuretic, confirm adherence and out-of-office BP, then evaluate resistant/secondary causes.

III. Hyperlipidemia and ASCVD Risk Reduction

Hyperlipidemia is best understood as a risk marker and a modifiable cause of atherosclerosis. The older dyslipidemia syndromes are still useful for pattern recognition, but modern clinic decisions are driven by ASCVD status, LDL-C level, diabetes, age, and estimated 10-year risk. The primary question is not simply “Is the cholesterol abnormal?” but “Will lipid-lowering therapy meaningfully reduce this patient’s future MI, stroke, or cardiovascular death?”

Lipid panel interpretation

Lipid value	Interpretation	Management implication
LDL-C	Most directly linked to ASCVD risk and statin benefit.	LDL-C >=190 mg/dL usually triggers high-intensity statin and evaluation for familial hypercholesterolemia.
HDL-C	Low HDL is a risk marker; high HDL is not a treatment target.	Exercise, smoking cessation, weight loss improve HDL but medications solely to raise HDL have not reliably improved outcomes.
Triglycerides	Elevated in obesity, diabetes, alcohol use, hypothyroidism, CKD, medications, familial disorders.	TG >=500 mg/dL increases pancreatitis concern; >=1000 mg/dL is high risk and needs urgent dietary/drug attention.
Non-HDL-C	Total cholesterol minus HDL; captures cholesterol in all atherogenic particles.	Useful when TG elevated.
ApoB / Lp(a)	Optional risk-enhancing markers in selected patients.	May clarify risk when family history or premature ASCVD is present.

Common secondary causes of dyslipidemia

• Hypothyroidism can raise LDL-C and should be considered when LDL is unexpectedly high or statin intolerance/myopathy is suspected.

• Diabetes, obesity, alcohol use, chronic kidney disease, nephrotic syndrome, and liver disease commonly raise triglycerides and/or alter lipoprotein patterns.

• Medications that may worsen lipids include glucocorticoids, estrogens, thiazides, nonselective beta-blockers, retinoids, protease inhibitors, atypical antipsychotics, and some immunosuppressants.

• Pregnancy can raise triglycerides; statin decisions must account for pregnancy status and reproductive planning.

Familial and classic dyslipidemia patterns

Pattern	Dominant abnormality	Clinical clues
Familial hypercholesterolemia	Very high LDL-C, often >=190 mg/dL in adults.	Tendon xanthomas, premature CAD, family history; treat aggressively.
Familial combined hyperlipidemia	LDL and/or TG elevated; apoB often high.	Common inherited pattern; premature ASCVD risk.
Familial dysbetalipoproteinemia	Remnant particles/IDL; cholesterol and TG both elevated.	Palmar xanthomas, tuberoeruptive xanthomas; often needs secondary trigger.
Severe hypertriglyceridemia	TG often >=500-1000 mg/dL.	Pancreatitis risk; eruptive xanthomas, lipemia retinalis at very high levels.

Statin benefit groups and intensity

Patient group	Typical recommended approach	Notes
Clinical ASCVD	High-intensity statin or maximally tolerated statin; add nonstatin therapy if LDL-C remains above threshold in high-risk patients.	Secondary prevention: prior MI, ACS, stroke/TIA of atherosclerotic origin, PAD, revascularization.
LDL-C >=190 mg/dL	High-intensity statin regardless of calculated 10-year risk.	Consider familial hypercholesterolemia and cascade screening.
Diabetes age 40-75	At least moderate-intensity statin; high-intensity if multiple risk factors or age 50-75.	Do not rely only on calculated risk.

Patient group	Typical recommended approach	Notes
Primary prevention age 40-75, LDL 70-189	Use 10-year ASCVD risk plus risk enhancers. Moderate-intensity generally when risk >=7.5%; high-intensity often when risk >=20%.	Shared decision-making: benefit, adverse effects, cost, preferences.
USPSTF primary prevention framing	Initiate statin age 40-75 with >=1 CVD risk factor and 10-year risk >=10%; selectively offer at 7.5% to <10%.	Applies to adults without ASCVD and LDL-C <190 mg/dL.

Intensity	Expected LDL-C reduction	Examples
High-intensity	>=50%	Atorvastatin 40-80 mg; rosuvastatin 20-40 mg.
Moderate-intensity	30% to <50%	Atorvastatin 10-20 mg; rosuvastatin 5-10 mg; simvastatin 20-40 mg; pravastatin 40-80 mg; lovastatin 40 mg.
Low-intensity	<30%	Simvastatin 10 mg; pravastatin 10-20 mg; lovastatin 20 mg.

Hypertriglyceridemia management

Triglyceride management has two goals: ASCVD prevention and pancreatitis prevention. Mild to moderate elevations often improve with weight loss, glycemic control, reduced alcohol intake, exercise, and treatment of secondary causes. When triglycerides are >=500 mg/dL, pancreatitis prevention becomes a priority; when >=1000 mg/dL, very-low-fat diet, elimination of alcohol/simple sugars, diabetes control, and drug therapy are urgent. Fibrates and prescription omega-3 fatty acids are used mainly for severe hypertriglyceridemia, while statins remain foundational when ASCVD risk is high.

Lipid follow-up and safety monitoring

• Obtain a baseline lipid panel before therapy and repeat approximately 4-12 weeks after statin initiation or dose change; then monitor every 3-12 months depending on risk and adherence questions.

• Baseline ALT is reasonable before starting statins. Routine repeated liver enzyme testing is not needed unless symptoms or clinical concern develops.

• CK is not needed routinely, but check it when severe muscle symptoms, weakness, dark urine, or high-risk drug interactions occur.

• Statin muscle symptoms require a careful history: distribution, timing, exertional injury, hypothyroidism, vitamin D deficiency, drug interactions, and rechallenge response.

Aspirin and other prevention decisions

Primary-prevention aspirin is no longer a routine default. For adults 40-59 years with a 10-year CVD risk of at least 10%, the net benefit is small and individualized; bleeding risk, patient values, and competing prevention options matter. New aspirin for primary prevention is generally avoided in adults 60 years or older because bleeding risk usually outweighs benefit. This does not apply to secondary prevention after established ASCVD, where antiplatelet therapy often remains indicated.

IV. Outpatient Headache: Triage Before Classification

Headache is common in ambulatory medicine, but the first step is not naming the primary headache syndrome. The first step is deciding whether the headache could be secondary and dangerous. Most chronic or recurrent headaches in clinic are migraine, tension-type headache, medication-overuse headache, or cluster headache. However, a small number reflect subarachnoid hemorrhage, meningitis, encephalitis, intracranial mass, temporal arteritis, venous sinus thrombosis, hypertensive emergency, glaucoma, carbon monoxide poisoning, pregnancy-related disease, or medication/toxin effects.

Headache red flags

Red flags include sudden thunderclap onset; “worst headache of life”; new neurologic deficit; papilledema; fever, meningismus, rash, or immunosuppression; cancer; pregnancy or postpartum state; age >50 at first onset; progressive pattern; positional headache; headache precipitated by exertion/sex/Valsalva; head trauma; anticoagulation; HIV or transplant status; and new headache with jaw claudication, scalp tenderness, or visual symptoms suggesting giant cell arteritis.

Red flag pattern	Concern	First diagnostic move
Thunderclap maximal in seconds-minutes	Subarachnoid hemorrhage, reversible cerebral vasoconstriction, dissection, venous thrombosis.	Noncontrast head CT urgently; LP/CTA/MRA depending on timing and suspicion.
Fever, neck stiffness, AMS, rash	Meningitis/encephalitis, sepsis.	ED evaluation, blood cultures/empiric therapy when appropriate, LP after imaging if indicated.
Age >50 with new headache	Giant cell arteritis, mass, subdural, vascular disease.	ESR/CRP if GCA symptoms; urgent steroids if visual symptoms; imaging based on features.
Papilledema or positional worsening	Raised intracranial pressure, mass, venous sinus thrombosis.	Neuroimaging; avoid LP until mass effect excluded.
Focal neurologic deficit or seizure	Stroke, mass, hemorrhage, infection.	Urgent neuroimaging and acute neurologic evaluation.
Cancer/immunosuppression/HIV	Metastasis, abscess, opportunistic infection.	MRI often preferred when stable; ED if acute or neurologic signs.

Primary headache pattern recognition

Syndrome	Typical features	Treatment anchors
Tension-type headache	Bilateral, pressing/tight band-like pain; mild-moderate; no vomiting; not worsened by routine activity; posterior neck/scalp muscle tenderness may occur.	Reassurance, trigger/sleep/stress management, NSAID or acetaminophen for episodic attacks; avoid medication overuse.
Migraine	Recurrent attacks lasting 4-72 hours; unilateral or bilateral throbbing pain, moderate-severe, worsened by activity, nausea/vomiting, photophobia/phonophobia; aura in some.	NSAID/acetaminophen early; triptan if moderate-severe and no contraindication; antiemetic; preventive therapy if frequent/disabling.
Cluster headache	Excruciating unilateral orbital/temporal pain, 15-180 minutes, with ipsilateral autonomic signs: tearing, conjunctival injection, rhinorrhea, ptosis/miosis, facial sweating; circadian clustering.	High-flow oxygen and subcutaneous/intranasal triptan for attacks; verapamil preventive; avoid alcohol during clusters.
Medication-overuse headache	Headache on >=15 days/month in patient using acute medications frequently; often superimposed on migraine or tension-type headache.	Educate, taper/withdraw overused medication, start preventive strategy when indicated.

Tension-type headache

Tension-type headache is often described as steady, aching, vice-like, or band-like pain encircling the head. It may be generalized or most intense in the neck, occiput, temples, or forehead, and it is commonly associated with posterior neck or scalp muscle tightness. It does not usually produce the prominent nausea, vomiting, photophobia, phonophobia, or activity intolerance typical of migraine. Management begins with excluding red flags, identifying sleep deprivation, stress, depression, anxiety, jaw clenching, cervical strain, visual strain, and medication overuse, then using limited NSAIDs or acetaminophen for episodic symptoms. Frequent analgesic use can worsen headache frequency.

Migraine bridge for ambulatory triage

Even when the source segment begins with tension and cluster headache, migraine must be included because it is frequently confused with both. Migraine is a neurovascular pain disorder with recurrent attacks, sensory sensitivity, and sometimes aura. A patient with unilateral throbbing pain, nausea, photophobia, phonophobia, and worsening with activity has migraine until proven otherwise. Triptans are useful abortive agents but are avoided in hemiplegic migraine, basilar/brainstem aura syndromes, uncontrolled hypertension, significant coronary/cerebrovascular disease, and high-risk vascular presentations. Preventive therapy is considered when attacks are frequent, prolonged, disabling, refractory to acute treatment, or acute medication use becomes excessive.

Cluster headache

Cluster headache is one of the trigeminal autonomic cephalalgias. Attacks are short compared with migraine, but the pain is severe and classically unilateral behind or around one eye. The patient is often restless rather than still. Attacks recur in clusters over weeks to months, often at the same time of day or night, and alcohol can trigger attacks during an active cluster period. The source chapter emphasizes middle-aged men, unilateral periorbital pain, ipsilateral lacrimation/flushing/nasal symptoms, 30-90 minute attacks, and remission periods; modern descriptions generally allow 15-180 minute attacks.

Cluster attack treatment	Details
Oxygen	100% oxygen by nonrebreather mask, commonly 12-15 L/min for about 15-20 minutes, works best when started early.
Triptan	Subcutaneous sumatriptan is fastest; intranasal triptans are alternatives. Avoid when vascular contraindications exist.
Prevention	Verapamil is first-line preventive; obtain ECG and monitor dose-related PR prolongation. Transitional steroids or occipital nerve block may be used while preventive therapy takes effect.

When to image headache in clinic

Stable recurrent headache with a normal neurologic exam and a clear primary headache pattern usually does not require neuroimaging. Image when red flags are present, the neurologic examination is abnormal, the pattern is new or changing, onset is after age 50, there is cancer or immunosuppression, the headache is thunderclap, posttraumatic, positional, exertional, or associated with papilledema, seizure, fever, or altered mental status. CT is favored for acute hemorrhage/trauma emergencies; MRI is generally more sensitive for mass, posterior fossa disease, inflammatory disease, and many subacute secondary causes.

Medication-overuse thresholds

Medication-overuse headache should be suspected when headache becomes frequent in a patient using acute medications repeatedly. Approximate risk thresholds are use on 10 or more days/month for triptans, ergotamines, opioids, or combination analgesics, and 15 or more days/month for simple NSAIDs or acetaminophen. The solution is not simply switching abortives; the patient needs education, withdrawal/taper strategy, treatment of the underlying primary headache, and a prevention plan.

V. Integrated Ambulatory Algorithms

Algorithm 1: Elevated BP found in clinic

Step 1: Repeat the measurement correctly and assess symptoms. Step 2: If neurologic deficit, chest pain, pulmonary edema, AKI, papilledema, aortic dissection symptoms, pregnancy emergency, or encephalopathy is present, send for emergency evaluation. Step 3: If severe but asymptomatic, check adherence/substances, arrange labs and close follow-up, and intensify oral therapy rather than rapidly lowering BP. Step 4: If stage 1 or stage 2, confirm with repeat visits and/or home BP, estimate ASCVD risk, evaluate target organs, start lifestyle for all, and start medication based on risk/stage. Step 5: Reassess within about 1 month after medication change and monitor electrolytes/creatinine when relevant.

Algorithm 2: Lipid panel in primary care

Step 1: Identify clinical ASCVD, diabetes, LDL-C >=190 mg/dL, TG >=500 mg/dL, and secondary causes. Step 2: For ASCVD, use high-intensity/maximally tolerated statin and consider LDL thresholds for nonstatin add-on in high-risk patients. Step 3: For LDL-C >=190 mg/dL, start high-intensity statin and evaluate familial hypercholesterolemia. Step 4: For diabetes age 40-75, prescribe at least moderate-intensity statin; consider high-intensity with added risk. Step 5: For primary prevention without those categories, calculate 10-year ASCVD risk and incorporate risk enhancers. Step 6: Recheck lipids 4-12 weeks after therapy changes.

Algorithm 3: Headache in ambulatory clinic

Step 1: Ask whether onset was sudden/maximal, whether this is the worst or first headache, and whether neurologic signs, fever, papilledema, trauma, cancer, immunosuppression, pregnancy/postpartum, anticoagulation, or age >50 at onset are present. Step 2: If red flags are present, select urgent imaging/lab/ED pathway rather than treating as primary headache. Step 3: If no red flags, classify by phenotype: migraine, tension-type, cluster, medication-overuse, sinus/ENT, ocular, cervicogenic, or other. Step 4: Treat acute symptoms, limit abortive frequency, address triggers/sleep/stress, and schedule follow-up if pattern is new, frequent, disabling, or diagnostically uncertain.

VI. High-Yield “Do Not Miss” Table

Finding	Why it matters	Action
BP >=180/120 + neurologic deficit, pulmonary edema, ACS symptoms, AKI, papilledema, dissection symptoms	Hypertensive emergency possible.	Emergency evaluation; IV therapy and organ-specific management.
ACE inhibitor + lip/tongue swelling	Bradykinin angioedema can obstruct airway.	Stop ACE inhibitor; airway-first evaluation; do not rechallenge.
LDL-C >=190 mg/dL	Possible familial hypercholesterolemia and high lifetime ASCVD risk.	High-intensity statin; family history/cascade screening; consider add-ons if response inadequate.
TG >=500 mg/dL, especially >=1000 mg/dL	Pancreatitis risk.	Address diabetes/alcohol/secondary causes; very-low-fat diet if severe; fibrate/omega-3 as appropriate.
New headache age >50 with jaw claudication or visual symptoms	Giant cell arteritis can cause irreversible blindness.	ESR/CRP; immediate glucocorticoids if suspected; temporal artery imaging/biopsy pathway.
Thunderclap headache	Subarachnoid hemorrhage and vascular emergencies.	Urgent CT and further evaluation if negative but suspicion persists.
Cluster headache phenotype	Highly treatable but often misdiagnosed as sinus or migraine.	High-flow O2 and triptan for attacks; verapamil prevention.

VII. Graph-RAG Concept Map

Concept node	Key edges for future retrieval
Hypertension	Causes target-organ damage; treated by lifestyle, thiazide, ACEi/ARB, CCB; complicated by stroke, CKD, LVH, CAD, HF, retinopathy; mimicked by white-coat HTN.
Secondary hypertension	Caused by renal artery stenosis, CKD, primary aldosteronism, OSA, pheochromocytoma, coarctation, drugs/substances; suggested by resistant or abrupt HTN.

Concept node	Key edges for future retrieval
Hyperlipidemia	Increases ASCVD risk; treated by statins; secondary causes include hypothyroidism, diabetes, nephrotic syndrome, CKD, liver disease, medications.
Hypertriglyceridemia	Causes pancreatitis when severe; associated with diabetes, alcohol, obesity, genetic syndromes; treated by secondary-cause correction, fibrates/omega-3 when severe.
Tension headache	Primary headache; distinguished from migraine by lack of nausea/activity worsening; treated with limited NSAID/acetaminophen and trigger management.
Cluster headache	Trigeminal autonomic cephalalgia; treated acutely with oxygen/triptan; prevented with verapamil; associated with unilateral autonomic signs.

References and source integration

This chapter segment was synthesized from the uploaded ambulatory medicine source pages covering hypertension, hyperlipidemia, and headache, plus current clinical guideline frameworks from ACC/AHA cholesterol and hypertension guidance, USPSTF statin and aspirin primary-prevention recommendations, and ACR headache imaging appropriateness guidance. The writing is original and designed for Medvale study use and future graph-RAG extraction.